How-To Library

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Handling Inflight UOWs Found at Log-Based ECCR Shutdown on z/OS

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Abstract

When you shut down a PowerExchange log-based ECCR on z/OS, it might report that an active UOW exists. Usually, the ECCR can handle this situation. However, if the UOW remains active in the PowerExchange Logger for MVS UOW connection, you need to take action. This article describes how to identify and handle inflight UOWs so that you can properly resume change data capture later. It focuses on the IMS log-based ECCR.

Supported Versions

- PowerExchange 10.0 or later

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Overview

After you issue a Close command to a PowerExchange log-based ECCR on z/OS, the ECCR might report that an active UOW exists. Usually, you do not need to take any action. During normal shutdown processing, the ECCR processes UOW records up to the last record before the PowerExchange Logger checkpoint or up to the record at which the UOW aborts. However, in certain environments, such as testing and development environments in which jobs are canceled often and IMS regions are canceled and cold started, some UOWs might remain active in the PowerExchange Logger for MVS UOW connection. This article describes how to identify these inflight UOWs and then determine whether to commit or abort them to maintain your change data capture (CDC) environment.

The article focuses on the IMS log-based ECCR on z/OS. However, the general strategy also pertains to the following PowerExchange log-based ECCRs on z/OS:

- Adabas log-based ECCR
- Datacom table-based ECCR
- IDMS log-based ECCR

Scenario

You want to determine if an inflight, or active, UOW exists before shutting down the IMS log-based ECCR.

To determine if an inflight UOW exists, you use one of the following methods:

- Look for message PWX-07291 in the message log:

```
PWX-06182 Totals   14/01/14 09:03:14          I=000000001735 U=000000000052
D=000000001299 C=000000000226 Log=14/01/13 18:31:52.05
PWX-07729 IMS MIGIMSC0, ECCR, Tue Jan 14 09:03:14.877 2014, Active unit of work found at
shutdown, UOW=C9E5C4F04040404000000144C00000000.
PWX-07291 ECCR is shutting down after reading the IMS log to 14/01/13 18:31:52.051355
```

- Issue the IMS log-based ECCR LIST OUOW command:

```
/F <eccr_name>,LIST OUOW
```

Note: This command is usually reserved for Informatica internal use only and is not documented in the *PowerExchange Command Reference*.

The command prints a PWX-06928 message for each open, or in-flight, UOW. The message includes the ECCR and IMS identifiers for the UOW. For example:

```
PWX-06928 MIGIMSC0 ECCR-UOW=MIGIMSC000000000000000DD4
IMSUOW=C9E5C4F04040404000000144C00000000 type=N
```

Task Flow for Handling Inflight UOWs

Perform the following high-level tasks to handle inflight UOWs in the PowerExchange Logger properly:

1. Verify that the ECCR processed all of the available database logs.
2. Display the UOWs that are in flight in the PowerExchange Logger UOW connection.
3. Determine the STCK value for the beginning of the inflight UOW.
4. Convert the STCK value into a timestamp in human-readable format.
5. Find the database log in which the UOW begins.
6. Locate the UOW start record in the log and find the IMS recovery token.
7. Analyze the UOW to determine whether to commit or abort it.
8. Commit or abort the UOW.

Step 1. Verify That the ECCR Processed All Available Logs

Verify that the ECCR has processed all of the available IMS logs.

Records for a transaction that are still in the online log data set (OLDS) are not available to the ECCR until an OLDS log switch occurs. A log switch can occur automatically in IMS or as a result of an IMS /SWITCH OLDS command.

Step 2. Display the UOWs That Are Active in the PowerExchange Logger UOW Connection

Enter the following PowerExchange Logger for MVS command to display message PWXEDM172679I, which presents the EDM Logger UOW CONNECTION report:

```
/F <logger_name>,DIS OBJECT=CONNECTION,TYPE=UOW
```

The following example report identifies the inflight UOWs for the IMS log-based ECCR named MIGIMSC0:

```
PWXEDM172679I EDM Logger UOW CONNECTION report follows:
               *ECCR          UOW              Type  Logger UOWid           Status  *
                MIGIMSC0 000000000000DD40000  UOW   MIGL0000001C6A360000   in-flight
                MIGIMSC0 FFFFFFFFFFFFFFFF0000  UOW   MIGL0000001CABA20000   in-flight
```

Note: The UOW identifier of FFFFFFFFFFFFFFFF0000 indicates a special UOW that is used for ECCR restart purposes. Use the other UOW identifier to identify the UOW that is in flight in the PowerExchange Logger UOW connection.

Step 3. Determine the STCK Value for the Beginning of the In-flight UOW

You can use the PowerExchange Log Scan and Print utility, EDMLUCTR, to get detailed information about the UOW, including the STCK time that corresponds to the beginning of the UOW.

Use the following JCL statements to run EDMLUCTR:

```
//READER   EXEC PGM=EDMLUCTR,PARM='TRACEE'
//STEPLIB  DD  DISP=SHR,DSN=PWX.Vxxx.LOAD
//         DD  DISP=SHR,DSN=PWX.VxxxDIST.LOAD
//EDMPARMS DD  DISP=SHR,DSN=ABCDEF.VTEST.ABCD.USERLIB
//ERDS01   DD  DISP=SHR,DSN=ABCDEF.VTEST.ABCD.ERDS01
//SYSPRINT DD   SYSOUT=*
//SYSUDUMP DD   SYSOUT=*
//SYSIN    DD   *
-SEL   LOGRBA=0000001C6A36 RECORDS=EOF PACKET-DETAIL
//*
```

Include the -SEL statement with the LOGRBA and PACKET-DETAIL parameters. The PACKET-DETAIL parameter prints detailed information, in hexadecimal format, about the UOW records. The LOGRBA value limits the EDMLUCTR output to the log RBA of the in-flight UOW. For the LOGRBA value, use the middle portion of the **Logger UOWid** value from the EDM Logger UOW CONNECTION report in Step 2. Omit the initial four letters and ending four zeroes. In this example, the value is 0000001C6A36.

When you submit the JCL, the utility prints the packet-detail report. For example:

```
10:26:00.74               Beg-pkt  EDP-UOW=MIGL  0000001C6A3600000000 ECCR-UOW=MIGIMSC00000000000000DD4
Logrba=0000001C6A360000
10:26:00.74                        Timestamp-16:02:34:08 Date-01/03/2014
10:26:00.74                 0000  00000003 00000014 00000088 00000088            h h
10:26:00.74                 0010  0000016A 00740101 D7E54088 00000000    ¦    PV h
10:26:00.74                 0020  E2E30000 00000000 00000000 0DD40000 ST           M
10:26:00.74                 0030  00000000 00000000 8000CC82 35280622
öb
10:26:00.74                 0040  7F28D4C9 C7C9D4E2 C3F0D4C9 C7D34040 " MIGIMSC0MIGL
10:26:00.74                 0050  0000001C 6A360000 00000000 001C6A36     ¦         ¦
10:26:00.74                 0060  00000000 00000000 D4C9C7C9 D4E2C3F0         MIGIMSC0
10:26:00.74                 0070  00000000 00000DD4 C4E3C8D6 D4C1E240        ABCDEFGH
10:26:00.74                 0080  1A35BD6C 00F87100 00E20000 01000000   ]% 8   S
10:26:00.74                 0090  00000000 00000000 00000000 00000000
10:26:00.74                 00A0  00000000 00000000 00000000 00000000
10:26:00.74                 00B0  00000000 00000000 00000000 00000000
10:26:00.74                 00C0  00000000 00000000 00000000 00000000
10:26:00.74                 00D0  00000000 00000000 00000000 00000000
10:26:00.74                 00E0  00000000 00000000 00000000 00000000
10:26:00.74                 00F0  00000000 00000000 00000000 00000000
10:26:00.74                 00F0  00000000 00000000 00000000 00000000
10:26:00.74                 0100  00000000 00000000 00000000 00000000
10:26:00.74                 0110  00000000 00000000 00000000 00000000
10:26:00.74                 0120  00000000 00000000 00000000 003C0000
10:26:00.74                 0130  0000D540 0000C9E5 C4F04040 40400000   N   IVD0
10:26:00.74                 0140  144C0000 00000000 0000CC82 35280622  <        öb
10:26:00.74                 0150  7F280000 00000002 9C7DCC82 35280622 "        'öb
10:26:00.74                 0160  7F280000 00000000 00000000 00000000 "
10:26:00.74               Chg-rec  EDP-UOW=MIGL  0000001C6A3600000000 ECCR-UOW=MIGIMSC00000000000000DD4
Logrba=0000001C6BA00000
10:26:00.74                        Source=VSM Func=UPDT Srcname=IMLIVD0d002dl1100000
10:26:00.74                        Timestamp-16:02:34:08 Date-01/03/2014
10:26:00.74               Chg-rec  EDP-UOW=MIGL  0000001C6A3600000000 ECCR-UOW=MIGIMSC00000000000000DD4
Logrba=0000001C6F440000
10:26:00.74                        Source=VSM Func=UPDT Srcname=IMLIVD0d002dl1100000
10:26:00.74                        Timestamp-16:02:35:58 Date-01/03/2014
10:26:00.74               Chg-rec  EDP-UOW=MIGL  0000001C6A3600000000 ECCR-UOW=MIGIMSC00000000000000DD4
Logrba=0000001C72FC0000
10:26:00.74                        Source=VSM Func=UPDT Srcname=IMLIVD0d002dl1100000
10:26:00.74                        Timestamp-16:02:36:13 Date-01/03/2014
```

From this report, you can determine the STCK value of the UOW start time. The STCK value always starts at x'14A' and is 8 bytes in length. In this example, the STCK value is CC82352806227F28 and is derived from the highlighted values shown on the lines that begin with 140 and 150, which are hexadecimal record offsets.

Step 4. Convert the STCK Value to a Timestamp in Human-Readable Format

Convert the STCK value from Step 3 to a human-readable date and time format.

To convert the STCK value, use an available conversion tool or command. You can use the REXX exec in the TOD member of the SYS1.SBLSCLI0 data set, which belongs to the MVS Interactive Problem Control System (IPCS). For example, the following exec statement converts the STCK value of CC82352806227F28:

```
exec 'sys1.sclscli0(tod)' 'CC82352806227F28'
CC823528 06227F28 : 2014/01/03 16:02:34.083879  TIMEZONE: 0000000000000000
```

To use the timestamp in Step 5 to locate the IMS log in which the UOW starts, convert the human-readable date to a three-digit Julian date.

Step 5. Find the IMS Log in Which the UOW Starts

For an IMS source, you can find the SLDS log in which the UOW starts in one of the following ways:

- Look at the names of the SLDS logs. In the names, the D*xxxxx* and T*xxxxxxx* values specify the log start date (D) and time (T). The date is in Julian format. Compare this date and time to the UOW start timestamp from Step 4. When you find the two logs that bound the UOW start timestamp, use the earlier one. For example, if the UOW start timestamp is 2014/01/03 16:02:34.083879, the UOW begins in the first of the following two logs:

  ```
  IMS1300.SLDSP.IVD0.D13352.T1308159.V0D          << The UOW starts in this log.
  IMS1300.SLDSP.IVD0.D14003.T2228590.V0D
  ```

- Run the DBRC LIST.RECON command. Then review the (START,STOP) timestamps in the output to find the SLDS log that includes the UOW start timestamp. For example:

  ```
  DSN=IMS1300.SLDSP.IVD0.D13352.T1308159.V0D              UNIT=SYSALLDA
   START = 13.352 13:08:15.913717          FIRST DS LSN= 00000000000282E5
   STOP  = 14.003 22:28:58.992641          LAST  DS LSN= 0000000000029C9C
   FILE SEQ=0001    #VOLUMES=0001
   CHECKPOINT TYPES=00: SIMPLE=N SNAPQ=N DUMPQ=N PURGE=N FREEZE=N
  ```

Step 6. Locate the UOW Start in the IMS log and Find the IMS Recovery Token

Locate the beginning of the UOW in the IMS SLDS log that was identified in Step 5. Then find the recovery token for that log record.

Use one of the following methods:

- View the SLDS data set in ISPF with HEX ON. Then issue the Find command with the STCK value from Step 3. For example, the following Find command uses the hexadecimal STCK value of CC82352806227F28:

  ```
  VIEW       IMS1300.SLDSP.IVD0.D13352.T1308159.V0D
  Command ===> f x'CC82352806227F28'
  ```

  The end of the log record is displayed. Scroll left to the beginning of the record. Then find the recovery token, which is always in the same position in a log record. The following example shows the beginning portion of the log record with the recovery token values highlighted:

  ```
  r...IVD0    ...<....öb....c.T\.à..öb....DTLPB05         DTHOMAS ...íØh.h .DTLD00
  9000CECF444400140000C8320280EE0400C83202CEDDCFF444444444CECDDCE40205880841CEDCFF
  940795400000004C0000C2586234300410C2584F43372050000000004386412000050800800433400
  ```

  Note: "IVD0" corresponds to the value C9E5C4F0 in the sample packet-detail report in Step 2. (See the lines that begin with 130 and 140.) This hexadecimal value represents the IMS ID.

- Issue the Find command with the UOW ID that was reported in the PWX-07729 message at ECCR shutdown or that was included the EDMLUCTR utility packet-detail report from Step 2.

  In this example, the UOW ID is C9E5C4F0404040400000144C00000000. In the EDMLUCTR utility packet-detail report from Step 2, the following highlighted values on the lines that begin with the 130 and 140 values show the hexadecimal UOW ID:

```
10:26:00.74                  0130  0000D540 0000C9E5 C4F04040 40400000   N    IVD0
10:26:00.74                  0140  144C0000 00000000 0000CC82 35280622  <         öb
```

  View the Find command output with HEX ON. Find the 16-byte IMS RTKN recovery token for the UOW, which is always at the same location. In the following example output, the recovery token value is represented by the highlighted values:

```
VIEW       IMS1300.SLDSP.IVD0.D13352.T1308159.V0D
Command =-> f x'C9E5C4F0404040400000144C00000000'
006553 r   IVD0       <    öb    c T\ à  öb    DTLPB05         ABCDEFG    íØh h  DTLD002     SEG1d   @  REPLREPLöb
       9000CECF444400140000C8320280EE0400C83202CEDDCFF444444444CECDDCE40205880841CEDCFFF40010ECCF8410700DCDDDCDDC832
       940795400000004C0000C2586234300410C2584F43372050000000004386412000050808004334002000002571404 2C2295739573C258
```

Step 7. Analyze the Inflight UOW

Analyze the IMS log records for the inflight UOW that is associated with the IMS RTKN recovery token to diagnose why the UOW was still active. For information about the IMS log records, see the IBM IMS documentation, more specifically Troubleshooting for IMS > Diagnosis > SYS - System service aids. Also, you can use the ILOGREC macro in the IMS MACLIB library to determine the individual log record DSECTs.

Step 8. Commit or Abort the Inflight UOW

After you analyze the inflight UOW, decide whether to commit or abort it.

To abort the UOW, issue the following PowerExchange Logger Resolve Indoubt (RI) command, where E is the ECCR name and U is the hexadecimal UOW ID from the PWXEDM172679I message in Step 2:

```
/f logger_name,RI E=MIGIMSC0,U=X'0000000000000DD40000',ACTION=A
```

If you want to abort the UOW, enter ACTION=A. If you want to commit the UOW, enter ACTION=C. For more information about this command, see the *PowerExchange Command Reference*.

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